SCHOOL OF COMPUTING
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
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(Winter Semester)

Course code: 10214AD701 **Course Name: Major Project**

TTS NO	Supervisor Name	VTU NO.	Student Name1	VTU NO.	Student Name 2
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Title of the Project: : Climate Analyzer: Deep Learning for Vegetation Stress Detection and Short-Term Risk Forecasting

Problem Statement

Climate change is causing irregular rainfall and temperature variations, affecting vegetation health and agricultural productivity. Existing systems analyze either satellite data or climate data separately, leading to inaccurate risk prediction. Hence, there is a need for an integrated AI-based system that combines satellite imagery and climate data for early detection of vegetation stress and short-term risk forecasting.

Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Targeted WKs (wherever applicable)	Justification (as applicable)
WP1	Depth of Knowledge	Cannot be resolved without in-depth engineering knowledge at the level of K3, K4, K5, K6 or K8	WK3, WK4, WK5, WK6, WK8	The project requires knowledge of NDVI calculation from Sentinel-2 data, CNN for pixel-level stress detection, and LSTM for forecasting rainfall and temperature. It also involves combining image-based and time-series outputs into a single risk system.

WP2	Range of conflicting requirements	Involve wide-ranging and/or conflicting technical and non-technical issues	WK4, WK5, WK6, WK8	The system balances vegetation stress detection from satellite images and short-term climate forecasting. It must handle trade-offs between spatial analysis (CNN) and temporal prediction (LSTM) while keeping the model lightweight and practical.
WP3	Depth of analysis required	Have no obvious solution and require originality in analysis	WK3, WK4, WK5, WK8	There is no direct method to combine NDVI-based stress detection and climate forecasting. The project involves defining NDVI anomaly (ΔNDVI), selecting a threshold (−0.10), and designing a rule-based fusion for risk classification.
WP4	Familiarity of issues	Involve infrequently encountered or novel problems	WK4, WK8	The project combines Sentinel-2 imagery with ERA5 climate data and applies both CNN and LSTM models, which is not a routine problem and involves integrating different types of data and models.
WP5	Extent of applicable codes	Address problems not encompassed by standards and codes of practice	WK6, WK8	The project does not follow fixed standards for vegetation stress detection. NDVI anomaly calculation and the −0.10 threshold are defined based on observation and experimentation, and risk rules are manually designed.
WP6	Stakeholder involvement and conflicting requirements	Involve collaboration across disciplines and diverse stakeholders	WK5, WK6, WK7	The system is useful for farmers, planners, and local authorities who need early information about vegetation stress and climate risks for planning and decision-making.

WP7	Interdependence	Address high-level problems requiring a systems approach	WK3, WK5, WK6	The system includes multiple connected modules such as NDVI calculation, CNN-based stress mapping, LSTM forecasting, and rule-based fusion. Each module depends on the others to produce final risk outputs.

Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Diverse resources and technologies	The project uses publicly available datasets such as Sentinel-2 satellite imagery and ERA5-Land climate data, along with deep learning models like CNN and LSTM for analysis and forecasting.
EA2	Level of Interactions	Optimization of complex interactions	The system integrates multiple components including image-based stress detection using CNN and time-series forecasting using LSTM. These outputs are combined through a rule-based fusion to generate final risk levels.
EA3	Innovation	Creative and research-based solutions	The project introduces a self-labeling approach using NDVI anomaly (ΔNDVI) and combines spatial and temporal models (CNN + LSTM) to produce vegetation risk maps, which improves over using a single data source.
EA4	Consequences to Society and Environment	Significant societal impact	The system helps in identifying vegetation stress and predicting climate risks, which can support agriculture planning and early decision-making, especially in areas affected by climate variability.
EA5	Familiarity	Extends beyond prior experience	The project applies deep learning models to satellite and climate data, which goes beyond routine engineering tasks and involves handling real-world environmental data.

Sustainable Development Goals (SDG) Mapping

SDG No.	SDG Title	Relevance to the Project
SDG 13	Climate Action	The system analyzes climate data such as rainfall and temperature and predicts future trends, helping in understanding and responding to climate variability and its impact on vegetation.
SDG 2	Zero Hunger	The project supports agriculture by detecting vegetation stress early using satellite data and forecasting short-term climate risks, helping farmers take timely actions to reduce crop loss.

SDG No.	SDG Title	Relevance to the Project
SDG 15	Life on Land	The project monitors vegetation health using NDVI and identifies stressed regions, contributing to better management and protection of land ecosystems.

Overall Justification: The project satisfies the characteristics of Complex Engineering Problems and Activities by requiring integration of satellite imagery and climate data, application of deep learning models (CNN and LSTM), and development of a fusion-based risk prediction system. It involves multidisciplinary knowledge, addresses real-world environmental challenges, and supports decision-making in agriculture and climate monitoring.

Depth of Knowledge – Required Knowledge Levels (K3–K8)

The problem addressed in this project cannot be resolved without in-depth engineering knowledge spanning the following **Knowledge Profile levels**:

K3 – Engineering Fundamentals

• Understanding of NDVI computation using satellite bands (NIR and Red) and basic image processing techniques for vegetation analysis.

• Knowledge of data preprocessing, monthly alignment, and handling time-series data for climate variables like rainfall and temperature.

K4 – Specialist Knowledge

• Knowledge of Sentinel-2 satellite imagery processing and ERA5-Land climate datasets for environmental monitoring.

• Understanding of deep learning models such as CNN for spatial stress detection and LSTM for time-series forecasting.

K5 – Engineering Design

• Design of a system that integrates NDVI-based vegetation stress detection with short-term climate forecasting.

• Development of a rule-based fusion approach to classify vegetation risk into low, medium, and high categories.

K6 – Engineering Practice

• Implementation of NDVI calculation, cloud masking, data preprocessing, and execution of CNN and LSTM models.

• Generation of outputs such as stress maps, forecast trends, and risk visualizations using real-world datasets.

K7 – Engineering Comprehension

• Ability to interpret NDVI anomalies and identify vegetation stress patterns from satellite imagery.

• Understanding and analyzing model outputs and climate trends to derive meaningful risk insights**.**

K8 – Research Literature and Emerging Technologies

• Use of concepts from recent research on NDVI-based vegetation monitoring and anomaly detection techniques.

• Application of CNN–LSTM based approaches for integrating spatial and temporal data in environmental prediction.

Signature

(Supervisor Name)